# EXPERIMENT 15
# Probabilistic Context-Free Grammar Parsing

## Aim

To parse a sentence using a probabilistic context-free grammar and select a likely parse.

## Procedure

1. Install NLTK if required.
2. Define a probabilistic context-free grammar with probabilities.
3. Create a Viterbi parser.
4. Parse the input sentence.
5. Display the parse tree and probability.

## Program

```
# Install first: pip install nltk
from nltk import PCFG
from nltk.parse import ViterbiParser

grammar = PCFG.fromstring("""
S -> NP VP [1.0]
NP -> Det N [0.7] | N [0.3]
VP -> V [1.0]
Det -> 'the' [1.0]
N -> 'student' [0.6] | 'teacher' [0.4]
V -> 'studies' [1.0]
""")
parser = ViterbiParser(grammar)
for tree in parser.parse("the student studies".split()):
    print(tree)
    print("Probability:", tree.prob())
```

## Output

```
OUTPUT — EXPERIMENT 15
(S (NP (Det the) (N student)) (VP (V studies)))
Probability: 0.42
```

## Result

Thus, the Viterbi parser found a likely parse using the probabilities in the PCFG.